MILITARY SPECIFICATION

WIRE, STEEL, CARBON, HIGH STRENGTH

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1 SCOPE

1.1 This specification covers high strength steel round wire.

1.2 CLASSIFICATION

1.2.1 SIZE.-- Wire shall be furnished in a size indicated in Table II, as specified.

2 APPLICABLE PUBLICATIONS

The following standards and publications of the issue in effect on the date of invitation for bids, forms a part of this specification to the extent specified herein:

2.1 SPECIFICATIONS

Federal

UU-P-271	Paper, Wrapping, Waterproof, Kraft.

Military

MIL-B-121	Barrier Material, Greaseproof, Flexible.

2.2 STANDARDS

Federal

Fed. Test Method Std. No. 151	Metals, Test Methods.

Military

MIL-STD-163	Steel Mill Products, Preparation for Shipment and Storage.

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(When requesting any of the applicable documents, refer to both title and number. All requests should be made via the cognizant government inspector. Copies of this specification and other unclassified specifications and drawings required by contractors in connection with specific procurement functions should be obtained upon application to the Commanding Officer, Naval Supply Depot (Code DSC), 5801 Tabor Avenue, Philadelphia 20, Pennsylvania. All other documents should be obtained from the procuring activity or as directed by the contracting officer.)

ASTM STANDARD

Method A318-56 Standard Method of Testing of Steel Spring Wire.

(Copies of ASTM standards may be obtained from the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pennsylvania).

3 REQUIREMENTS

3.1 CHEMICAL COMPOSITION. -- The chemical composition shall conform with the respective limits of Table I. Individual determinations may vary from the specified range to the extent indicated in the check analysis column.

3.2 DIMENSIONS

3.2.1 DIAMETRAL TOLERANCES. -- The permissible variation from the nominal diameter of wire 0.030 inch and less in diameter is 0.0005 inches. On wire larger than 0.030 inch diameter, the tolerance is 0.001 inch. The tolerance on out-of-roundness shall be half the diametral tolerance.

3.3 MECHANICAL PROPERTIES

3.3.1 TORSIONAL TENSILE STRENGTH DUCTILITY.-- The torsion, ductility, and tensile breaking strength minimum requirements are indicated by Table II for various wire diameters when tested as indicated in 4.5.2, 4.6.2 and 4.7.2 respectively.

3.4 JOINTS. -- Wire rods and wires may be joined by welding provided such operations are accomplished in the initial stages of processing and that microstructural changes due to welding are eliminated by repeated annealing and drawing operations.

3.5 WORKMANSHIP.-- Wire shall be free from defects such as seams, pits, die marks, scrapes, cold shuts and other material or surface imperfections tending to impair the serviceability of the wire. Wire surfaces shall be smooth.

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TABLE I

CHEMICAL COMPOSITION

Element	Range, Per cent	Check Analysis Tolerance, Per cent
Carbon	0.85--0.90	0.02
Manganese	0.25--0.40	0.02
Silicon	0.15--0.25	0.02
Phosphorous	0.025 max	0.005
Sulfur	0.020 max	0.005
Chromium	0.04 max	
Molybdenum	0.04 max	
Copper	0.08 max	
Nickel	0.03 max	

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TABLE II

DUCTILITY, TORSION, AND TENSILE STRENGTH REQUIREMENTS

Wire Diameter, Inch	Torsion, Number of 360° Twists in 8"	Ductility Test		Tensile Breaking Strength	
		Sheave Diameter Inch 2/	Sheave Center Spacing Inch 2/	Stress, 1000 psi	Load Pounds
0.004	--	----	------	575	7.0
0.005	--	----	------	555	10.8
0.006	--	----	------	540	15
0.008	--	----	------	525	26
0.010	90	----	------	500	39
0.012	75	----	------	475	53
0.014	60	----	------	450	69
0.016	55	----	------	450	90
0.018	53	----	------	400	102
0.020	50	----	------	400	125
0.022	48	----	------	400	152
0.024	45	----	------	400	180
0.025	43	0.866	1.0	390	195
0.028	40	0.866	1.0	390	240
0.032	36	0.866	1.0	375	302
0.036	32	1.181	1 3/8	370	376
0.040	28	1.181	1 3/8	370	465
0.042	27	1.181	1 3/8	365	505
0.045	25	1.181	1 3/8	365	587
0.048	24	1.181	1.3/8	345	624
0.050	23	1.181	1 3/8	345	690
0.051	22	1.181	1 3/8	340	690
0.054	21	1.181	1 3/8	340	778
0.056	20	1.457	1 5/8	335	824
0.058	20	1.457	1 5/8	335	885
0.062	18	1.457	1 5/8	325	997
0.064	17	1.457	1 5/8	325	1045
0.072	15	1.457	1 5/8	325	1322
0.076	15	----	------	320	1450
0.080	14	----	------	315	1610
0.085	13	----	------	315	1790
0.090	12	----	------	310	1970
0.095	11	----	------	310	2195
0.100	10	----	------	305	2400

2/ Tolerance on roll groove diameter plus .050 -0 inch. Tolerance on distance between centers, 1/8 inch.

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4 QUALITY ASSURANCE PROVISISONS

4.1 INSPECTION RESPONSIBILITY. -- The supplier is responsible for the performance of all quality assurance tests with the exception of tests on samples which may be selected by the procuring activity for referee tests in a government laboratory. The inspection services and facilities employed must be acceptable to the Government. Inspection records of examination and tests shall be kept complete and available to the Government for a period of two years from date of delivery of the wires represented. The Government reserves the right to perform any or all of the prescribed tests for compliance with the requirements specified herein.

4.2 GENERAL. -- All the tests of this specification are classed as inspection tests for which the necessary sampling techniques and methods of testing are specified in this section.

4.3 DIMENSIONAL VISUAL INSPECTION. -- Two determinations of wire diameter shall be made on the wire on each shipping reel. The exposed surfaces of wire shall be visually examined for compliance with the workmanship requirements while being wound on the shipping reels.

4.4 CHEMICAL COMPOSITION

4.4.1 SAMPLING. -- One sample for testing shall be selected to represent ten shipping reels of wire of one diameter and composition except that when wire supplied can be positively identified with the specific heat of steel from which the wire material was obtained, certified copies of the ladle analysis supplied by the steel producer will be acceptable.

4.4.2 METHOD. -- Tests shall be conducted in accordance with methods, 111.1 or 112.1 of FED TEST METHOD STANDARD NO. 151.

4.5 TORSION TEST

4.5.1 SAMPLING. -- One specimen shall be selected from the first, last, and one intermediate shipping reel from each manufacturing coil at the time the wire is being transferred to the shipping reels.

4.5.2 METHOD. -- Torsion test specimens shall not be less than 10 inches in length. One end of the specimen shall be clamped in the fixed head of the testing machine and the other end clamped in the movable head. The distance

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between the clamps in which the wire is held shall not be less than 8 nor more than $8\frac{1}{2}$ inches. The wire shall be twisted at a uniform rate, in one direction until failure occurs. The speed of the machine shall not exceed 33 r.p.m. for wire diameters 0.125 inch or less, or 20r.p.m. for wires of larger diameters. Sufficient tension shall be applied longitudinally to the wire to avoid kinking during the test.

4.6 DUCTILITY TEST

4.6.1 SAMPLING.-- All the wire supplied in accordance with this specification shall be tested.

4.6.2 METHOD.-- Wire shall be passed through a series of five rolls or sheaves as it is transferred from manufacturing coils or reels to shipping reels. The sheaves shall be of the size and spacing between centers of rotation indicated by Table II for the respective wire size. The sheaves shall be arranged as indicated by Figure 1. Breakage of the wire is cause for rejection. No joining or splicing or retest of shorter lengths is permitted.

4.7 TENSILE TEST

4.7.1 SAMPLING.-- One specimen shall be selected from the first, last, and one intermediate shipping reel from each manufacturing coil at the time the wire is being transferred to the shipping reels and subsequent to the ductility test of par. 4.6.

4.7.2 METHOD.-- Wire specimens shall be tested in accordance with Method A318-56, ASTM Standard Method of Tension Testing of Steel Spring Wire.

4.8 REJECTION AND RETEST.-- The failure of any specimen to meet any of the requirements specified herein shall be cause for rejection of the shipping reels from the manufacturing coil represented, except that the contractor may of his option test samples from the coils not previously sampled and disposition made on the basis of the test results.

5. PREPARATION FOR DELIVERY

5.1 PRESERVATION AND PACKAGING.-- Preservation and packaging shall be Level A or C as specified (see6.2).

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5.1.1 LEVEL A.-- The wire shall be preserved and packaged in accordance with the requirements and procedures for Level A as specified for wire, high carbon and special grades, and alloys in MIL-STD-163. Each coil or reel shall consist of one continuous length of wire. The wire shall be coated with P-2 or P-6 type preservation of Specification MIL-P-116 with a greaseproof wrap conforming to Specification MIL-B-121, grade A, over-wrapped with a layer of waterproof barrier material conforming to Specification UU-P-271 class B-2, B-3 or C-1.

5.1.2 LEVEL C.-- The wire shall be packaged to afford the minimum degree of protection necessary to prevent deterioration or damage during shipment under normal enviromental conditions and commercial modes of transportation.

5.2 PACKING.-- Packing shall be Level A, B, or C as specified (see 6.2).

5.2.1 LEVEL A.-- The wire preserved and packaged as described in 5.1 shall be packed in overseas exterior type container in accordance with the requirements and procedures as defined by Military Standard MIL-STD-163.

5.2.2 LEVEL B.-- The wire preserved and packaged as described in 5.1 shall be packed as in Level A, except that the exterior container shall be of domestic exterior type.

5.2.3 LEVEL C.-- The wire that require(s) overpacking by the carrier shall be packed in exterior type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery, and shall meet as a minimum, the requirements of the rules and regulations applicable to the mode of transportation selected.

5.3 MARKING.-- In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of Standard MIL-STD-129.

SPECIAL MARKING

"Shipping reels shall be identified with lettering or with aluminum marking plates securely attached to the ends of the reels and marked with the following information:

Purchase Order or Contract Number

Manufacturer's Name

Specification MIL-W (WEP)

Wire Diameter and Approximate Length."

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In addition, a tag with the above marking information shall be firmly affixed to the outside of the individual boxes or the exposed water-proof wrapping paper in such a manner that the information will be fully visible or reels stacked end to end.

6 NOTES

6.1 INTENDED USE.-- The wire is intended for use in applications which require high strength and uniform properties.

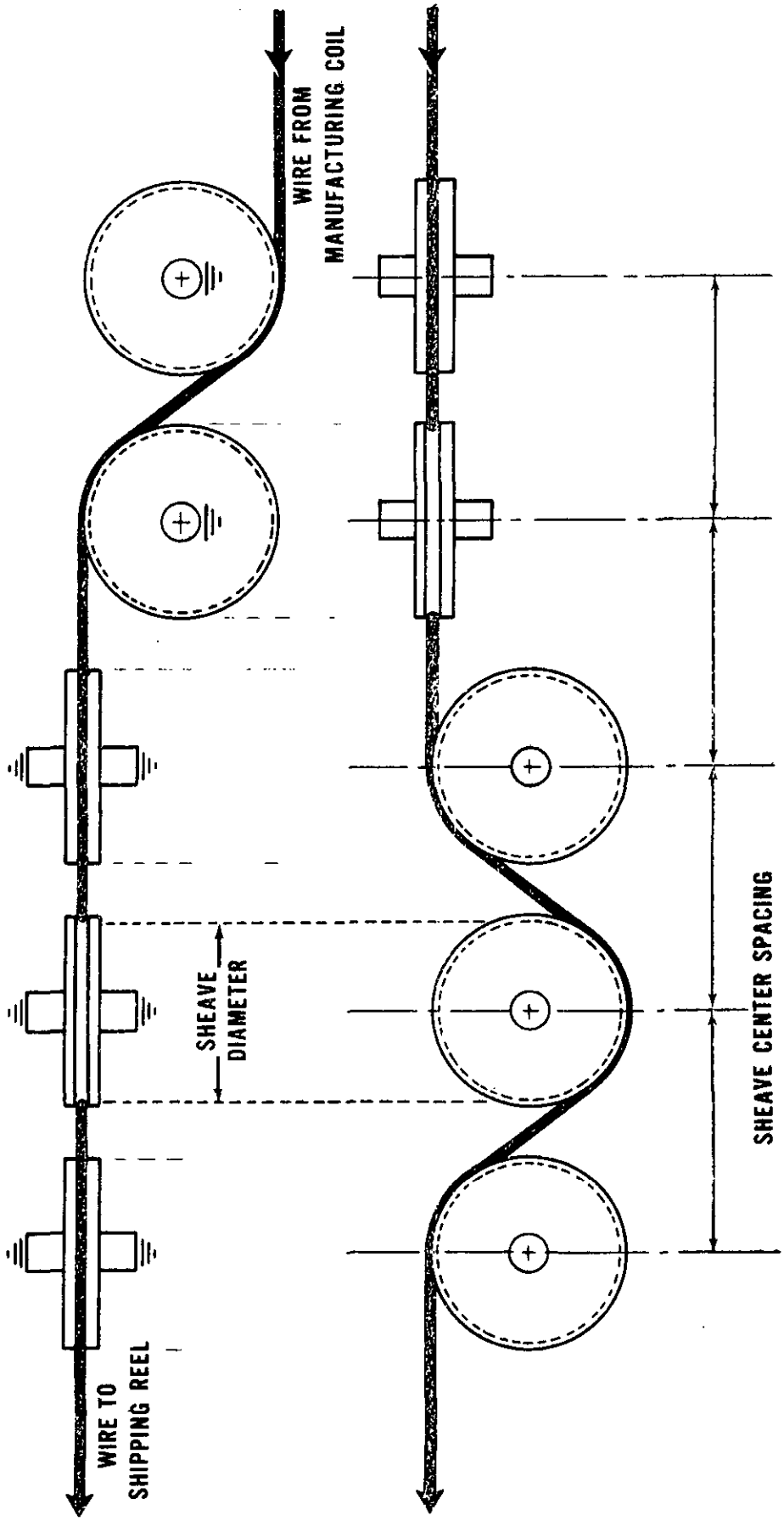
6.2 ORDERING DATA.-- Invitations to bid, purchase orders and contracts should specify the following:

(1) Wire diameter
(2) Amount of wire desired
(3) Length of wire, each coil or reel
(4) The level of packaging and packing desired

PATENT NOTICE.-- When the Government drawings, specifications, or other data used for any purpose other than in connection with a definetely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by imolication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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DUCTILITY TEST
Figure 1.



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